FOUNDATION COURSE

START YOUR
PREPARATION
FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 24

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Recap *of previous lecture*

AVERAGE & WEIGHTED AVERAGE



Topics *to be covered*

"SET THEORY"



SET:- "Collection of well defined objects is called Set"

eg Collection of all vowels in Eng alphabets ✓ i.e. $A = \{a, e, i, o, u\}$

eg " " 1st five prime nos ✓ i.e. $B = \{2, 3, 5, 7, 11\}$

eg " " all the states in INDIA ✓ i.e. $C = \{UP, B, H, D, \ldots\}$

eg " " good teachers in PW ✗

eg " " long Rivers in INDIA ✗

Null Set / Void Set / Empty Set $\rightarrow \phi = \{\ \}$

eg $A = \{x ; x \in N \ \& \ 5 < x < 6\} = \phi$ eg $B = \{x ; x \in R \ \& \ x^2 + 9 = 0\} = \phi$

Representation of set →

ROSTER Form	Set Builder Form
eg $A=\{1,2,3,4,5,6,7\}$ = Finite set,	$A=\{x;\ x\in N \text{ & } 1\le x\le 7\}$
eg $A=\{2,4,6,8,10,\dots\}$ = Infinite set	$A=\{x;\ x=2n \text{ where } n\in N\}$
eg $A=\{1,2,3,6,9,18\}$ = finite set	$A=\{x;\ x \text{ is divisor of } 18\}$
eg $A=\{1,\frac{1}{2},\frac{1}{3},\frac{1}{4},\dots\}$ = ∞ set	$A=\{x;\ x=\frac{1}{n} \text{ where } n\in N\}$

Universal Set → the set consist all the elements under consideration known as U-set

eg $A=\{1,2,3,5,7\}$, $B=\{2,3,8,9\}$, $C=\{6,4\}$ then $U=\{1,2,3,4,5,6,7,8,9\}$ = finite set

eg if we are dealing with Natural Nos then $U=N$

eg if we are studying 2-D Coordinate geometry then $U = ? = XY$ plane $= R \times R$

Cardinal Number → "Number of distinct elements in a set is called it's C. Number"

eg $A = \{a, e, i, o, u\} \Rightarrow n(A) = 5$

eg $A = \{x ; x$ is a letter from the word 'MATHEMATICS'$\}$

$= \{M, M, T, T, A, A, E, H, I, C, S\} = \{M, T, A, E, H, I, C, S\}$

$\Rightarrow n(A) = 8$ eg $\boxed{n(\phi) = 0}$

Subset → If every member of A is a member of B then A is called subset of B i.e $A \subseteq B$ & B is called superset of A i.e $B \supseteq A$

$\Rightarrow A \subset B$ & $B \supset A$

eg $A = \{1, 2, 3, 5\}$, $B = \{1, 2, 3, 4, 5, 6, 7\}$

eg if $A = \{a, b, c\}$ then write all the subsets of A ?

Sol: $\phi, \{a, b, c\}, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}$

Improper subset — Proper subsets

Note:-

(1) Any given set has exactly **two** Improper subsets one is ϕ & other is set itself.

(2) Null set is a subset of every set (T)

(3) Every set is a subset of itself (T)

(4) Total Number of Subsets of any Set with Cardinal number $N = 2^N$

Proof: Let $A = \{x_1, x_2, x_3, \ldots, x_N\} \Rightarrow n(A) = N$

So various subsets are $= {}^NC_0 + {}^NC_1 + {}^NC_2 + {}^NC_3 + \ldots + {}^NC_N$

$= (1+1)^N = 2^N$

eg if $n(A) = x$ & $n(B) = y$ & Total No. of subsets of A is 56 more than Total No. of subsets of B then find x & y ?

Sol:

$2^x = 2^y + 56$

$2^x - 2^y = 56$

$2^x - 2^y = 8 \times 7$

$2^y\left[2^{x-y} - 1\right] = 2^3(2^3 - 1)$

$y = 3$ & $x - y = 3 \Rightarrow x = 6$

Equal Sets :- Two sets A & B are called equal if

Every member of A is a member of B i.e $A \subseteq B$

& " " B " " " A i.e $B \subseteq A$

then $\boxed{A = B}$

Equivalent Sets :- A & B are called Equivalent sets if $n(A) = n(B)$

i.e if $[n(A) = n(B)]$ then $[A \sim B]$

eg if $A = \{1, 2, 3\}$ & $B = \{a, b, c\}$ then $A \sim B$ but $A \neq B$

eg if $A = \{1, 2, 5, 6\}$ & $B = \{5, 6, 1, 2\}$ then $A = B$

eg if $A = \{x ; x$ is a letter from the word 'ALLOY'$\} \Rightarrow A = \{A, L, O, Y\}$

& $B = \{x ;$ " " " " " 'LOYAL'$\} \Rightarrow B = \{L, O, Y, A\}$

$\Rightarrow \boxed{A = B}$

Power Set :- Collection of all the subsets of given set is called Power Set

eg $A=\{1,2,3\} \Rightarrow P(A)=\{\phi,\{1\},\{2\},\{3\},\{1,2\},\{2,3\},\{1,3\},\{1,2,3\}\}$

eg $A=\{a,b\} \Rightarrow P(A)=\{\phi,\{a\},\{b\},\{a,b\}\}$

Note :- Number of elements in a Power Set of given set = Total No. of subsets

if $n(A)=N$ then $n\{P(A)\}=2^N$

eg evaluate $n(P(P(P(\phi))))=?$

$\phi=\{\} \Rightarrow P(\phi)=\{\phi\}$

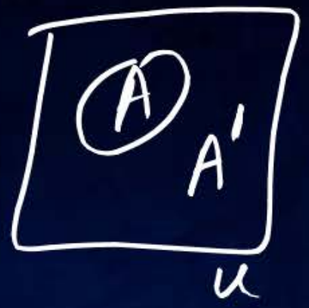
$\therefore P(P(\phi))=\{\phi,\{\phi\}\}$

$P(P(P(\phi)))=$

$=\{\phi,\{\phi\},\{\{\phi\}\},\{\phi,\{\phi\}\}\}$

$\Rightarrow n(P(P(P(\phi))))=4$

VENN-EULER Diagrams →

① 

; A' = Complement of $A = U - A$

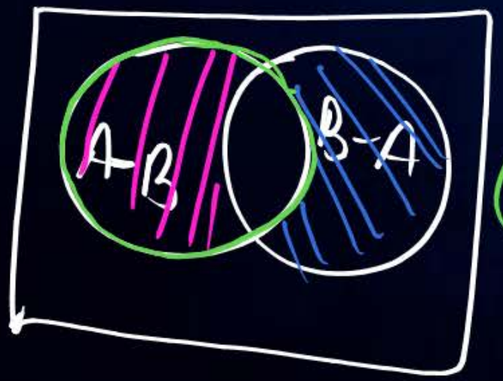


② $A \cup B = (A \cap B') \cup (A \cap B) \cup (A' \cap B)$

$= (A - B) \cup (A \cap B) \cup (B - A)$

$= U - (A' \cap B')$ = Either A or B or Both

③ Difference of Sets →



$A - B = A \cap B' = A - A \cap B$ = only A

$B - A = B \cap A' = B - A \cap B$ = only B

④ Symmetric Diff → $A \triangle B = (A - B) \cup (B - A) = (A \cup B) - (A \cap B)$

= exactly one of A or B

Law of Algebra of Sets →

(1) Idempotent Law: $A \cup A = A$ & $A \cap A = A$

(2) Identity Law: $A \cup \phi = A$ & $A \cap U = A$

ϕ is the identity element for union & U is the identity element for intersection

(3) Commutative Law → $A \cup B = B \cup A$ & $A \cap B = B \cap A$

(4) Associative Law → $A \cup (B \cup C) = (A \cup B) \cup C$ & $A \cap (B \cap C) = (A \cap B) \cap C$

(5) Distributive Law → $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

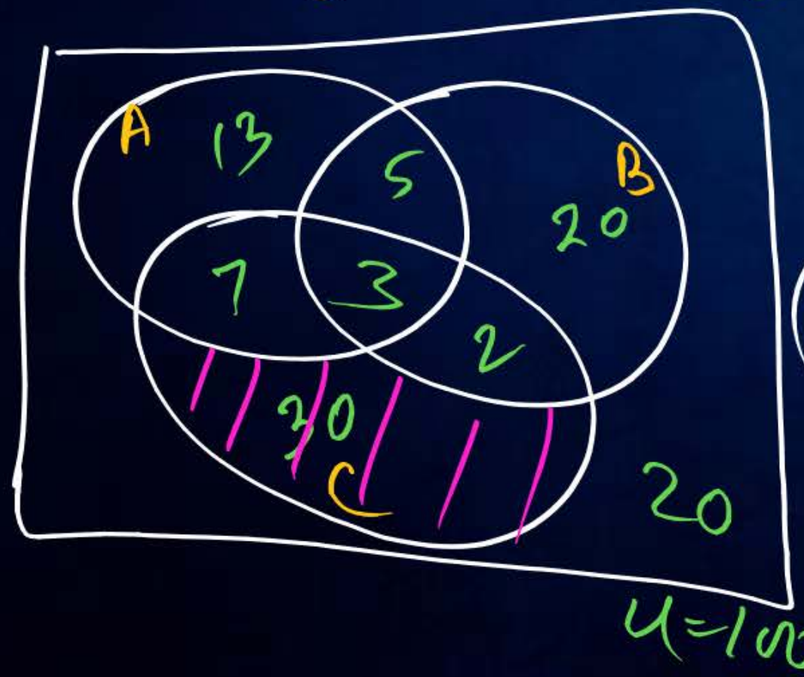
& $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

(6) De Morgan's Law → $(A \cup B)' = A' \cap B' =$ Neither A Nor B

& $(A \cap B)' = A' \cup B'$

Q. In a survey of 100 persons, it was found that 28 read Magazine A, 30 read B, 42 read C, 8 read A & B, 10 read A & C, 5 read B & C, & 3 persons read all three magazines then (i) How many read only C?
(ii) How many read none of the 3 Magazines?

Sol:



$n(A \cup B \cup C) = (13+20+30)+(5+7+2)+3$

$= 63+14+3 = 80$

(ii) $n(\bar{A} \cap \bar{B} \cap \bar{C}) = U - n(A \cup B \cup C) = 100-80$

$= 20$

(i) $n(\text{only } C) = 30$


Thank
YOU